

Greening the Chain: exploring the role of Supply Chain Management in advancing Corporate Sustainability. A dairy industry case study



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Introduction

Over the last few years, market demand in terms of sustainability has become increasingly pressing (Kumar, et al. 2012). Re-thinking and re-planning companies' activities from a sustainable perspective is no longer a voluntary choice but has rather become a duty: today companies are called upon to question their own environmental impact. Sustainability, intended not just in terms of environment, has thus become an important driver for all business decisions concerning supply chain activities and processes: today we call it **Sustainable Supply Chain (SSC)**, where transparency and sustainability are keywords and fundamental components of new business strategies (Belvedere e Grando 2017).

This new direction carries a twofold advantage for companies: first, it allows them to structure and manage a sustainable, efficient and responsive supply chain, as demanded by customers; secondly, it ensures a resilient supply chain network. As evidence of the rapid evolution of a market oriented towards sustainability, in just few years numerous international and national guidelines have been defined and implemented for measuring environmental performance (just to mention a few ones: the ISO 14001 standard, the EMAS scheme, the GRI 4.1 guidelines, the European Corporate Sustainability Reporting Directive, etc.).

While sustainability issues are growing in importance and are escalating the agendas of C-executives worldwide, the methodologies and approaches to manage them are just starting to develop and to be embedded along companies operational and strategic routines (León-Soriano, Muñoz-Torres e Chalmeta-Rosaleñ 2010).

One of the major challenges relies in **the alignment between traditional performance indicators** (such as profitability, productivity, service performance, quality, etc.) **with new indicators introduced by sustainability** related topics (e.g., carbon footprint indicators, LCA, scope 1-2-3 decomposition of company's footprint, resource consumption, circularity index, etc.). As most of the times, when trying to pursue different objectives on different performance (such as the one of sustainability and profitability) the outcome may result in a trade-off situation which may leave the company in a strategic impasse (Figge e Hahn 2012).

This situation highlights the emerging need for tools and methodologies able to support multi-performance decision making across company's levels in rapid times and with reliable results.

If on one hand there is a need for systems that could help managers and entrepreneurs in taking more informed decisions, on the other hand collecting data about the sustainability performance of a process may result in a burdensome activity due the general lack of data and cheap technologies (Al-Emran e Griffy-Brown 2023).

In order to solve this issue and to provide decision makers with a holistic overview on performances, **dynamic simulation modelling** appears to be very suitable for this purpose. The reason for adopting a quantitative technique such as simulation modelling, capable of describing the behavior of the analyzed variables over time, is simple: since environmental sustainability is measurable in quantitative terms, there





are no explicit reasons to exclude this area of analysis from the panel of performance simulated in a model or in a scenario.

Since the model is usually used to represent and simulate, with high accuracy, a specific company's supply chain and its related performance, it could be easily expanded also to track and estimate the environmental footprint of the supply chain itself, which is mainly caused by the production and transportation of goods across the value chain (Bové e Swartz 2016).

Once validated, the supply chain model (which under certain conditions may evolve in a supply chain digital twin) is then capable of estimating concurrently both financial, operational and sustainable performance of the value chain. This characteristic provides decision makers with punctual and reliable insights on what could be the real impacts, benefits and threats of implementing certain decisions, especially when trying to enhance the sustainability performance.

For a better understanding of this methodology, this article presents a real case application of simulation modelling for the deployment of the sustainability strategy of a company working in the dairy industry. After a short introduction which illustrates the main principles of dynamic simulation modelling and the reasons of its application to company's environmental sustainability issues, the second part of this article illustrates how it could be practically employed in the dairy industry case study. The detailed explanation of the case study will give the reader, hopefully, a clearer picture of what modelling sustainability entails and of what could be the benefits of adopting a quantitative approach when dealing with company's sustainability performance.





Simulation modelling for sustainability decision making

The rapid changes in the business environment push entrepreneurs and managers to continuous efforts in studying and overwhelming competitors to enhance company's market share. These ongoing changes result in an increasing complexity of the system and its underlying relationships in which new challenges arise, such as sustainability and social responsibility. In this sense, the use of simulation allows to face in a systemic manner typical supply chains challenges and suggests solutions that consider not only the traditional performance indicators but also new objectives, oriented towards sustainability.

Reality can therefore be modelled according to the desired level of detail using different variables: simulation succeeds in describing how these variables behave and interact in the course of time, empowering the understanding of their causal relations.

As an example, hereafter it is possible to list some action areas in which simulation can be applied effectively in the sustainability field when considering the supply chain perimeter:

- Production area: design and implementation of new production paradigms based on the principles of the circular economy, reduction of the amount of raw materials employed, re-use of scrap/by-products/waste, use of alternative materials with reduced environmental impact.
- Inventory area: inventory policies can be modified to leverage on stock replenishment parameters to achieve optimization in transportation resources utilization, by considering at the same time the costs and emissions deriving from the loading and unloading processes.
- Transportation area: distribution planning can be structured efficiently in order to determine which are the optimal volumes to be transported and/or which are the most suitable routes and means of transport to reduce CO2 emissions.
- Distribution area: redesign the distribution network to achieve a reduced environmental footprint.



The 3S approach: sustainability, supply chain & simulation

Given the premise of addressing the supply chain sustainability strategy development in a quantitative way, the author proposes an innovative methodology under the name of *3S Approach*. This methodology combines practical elements from the field of supply chain management with quantitative techniques, with the final aim of streamlining the sustainability performance of a company.

The pillars of the *3S Approach* are:

- **Supply chain** framed;
- **Sustainability** as focal performance;
- **Simulation modelling** as tool used for strategy analysis and development.

Starting from the first element, **supply chain** has been identified as major companies' area affecting the sustainability performance (Bové e Swartz 2016). Researchers show in fact that processes belonging to supply chain and operations areas are accountable on average for the 80% of total company emissions.

Focusing on supply chain is then crucial to address seriously the sustainability issue, since even a small enhancement could lead to important benefits for the environment. Additionally, supply chain represents, for most of the industries, the major area of value creation: this led during the last century to the development of several methodologies, approaches and practices with the aim of streamlining the operational performance and consequently the profitability and service level of a company. While some of these approaches focused mainly on a single performance (e.g. 6 Sigma for quality), others considered also some environmental aspects such as the generation and management of manufacturing waste and scraps (e.g. Lean manufacturing). The focus on supply chain excellence could then be enlarged to the totality of environmental performance, and most of the times this could lead to investigate directions of improvement that could be beneficial also for company profitability.

This last consideration introduces the second pillar of the *3S Approach*, the focus on the **sustainability performance**. It is opportune to state that, being an analytical and quantitative methodology, the *3S Approach* could be focused only on environmental sustainability and not on social sustainability, since the last cannot be modelled through analytical mindset.

Given this necessary premise, environmental sustainability can instead being practically measured and declined in its major components (EFFAS & DVFA 2010), such as: carbon footprint, harmful emissions (concerning water, air and soil), water usage, energy usage, renewable and non-renewable materials consumption, waste and scrap generation, circularity.

Measuring the different impacts on sustainability KPIs is then crucial for companies to understand the critical areas, to prioritize the interventions and finally to gain the major efficacy from their investments.





At the same time, sustainability cannot be considered as the unique objective performance when evaluating strategic improvements. The deployment of the sustainability strategy should in fact consider also the financial and operational implications which could affect the traditional performances of the company. For this reason, the need for a comprehensive approach, capable of estimating the impacts on all relevant performances, arises. Moreover, very few are the contexts in which sustainable and financial indicators are in a linear and well known relation: most of the times company's performance are on a trade-off relation and can be visualized as nonlinear relations, that the human mind is not capable to identify, solve or interpret correctly alone.

This last characteristic leads to the use of the third pillar, the one of **simulation modelling**. In fact, simulation is among the few technologies capable of representing the complexity of the observed reality, through a set of variables and interdependencies that constitutes the digital twin.

Simulation allows to easily include in the model different degrees of complexity, to investigate the relations among variables and to conduct the cause-effect analysis on the results provided. It opens the possibility to understand the behavior of complex systems as modern supply chains are and to discover the best direction of improvement by looking carefully and holistically at all the relevant performance indicators.

The advantages deriving from the *3S Approach* are numerous, among the most important there are:

- A better **representation of business reality**, thanks to the introduction of variables with stochastic behavior that can describe processes subject to great variability.
- An **in-depth analysis of risks and opportunities** linked to modifications of existing parameters and/or the introduction of new paradigms, enabled by the benchmark analysis conducted on the performances of different scenarios created through the simulation model.
- An **optimization of relevant parameters** to achieve solutions of local optimum valid only in the given context and, therefore, which are defined “ad hoc” according to the features of the modeled reality.
- A **systemic vision** of the actual network, that allows to capture the changes in business performances when the supply chain is subjected to improvement processes.





The dairy industry case study

The explanation of how the *3S Approach* could be practically used when dealing with supply chain sustainability is carried out in the following paragraphs through the description of a case study, created by a team of professional supply chain consultant and academic staff from the Economy faculty of the Catholic University of Milan (Italy).

The team decided to frame the case study on a simple and well know consumer product, the yogurt, since this industry has a great relevance in terms of carbon footprint and economic growth.

In fact, in Italy the food sector, which includes agriculture, food industry, distribution (including HORECA), is one of the leading economic sectors with revenues over 500 billion euros. At the same time, it is also one of the most polluting (Montanino, et al. 2021). An estimation suggests that, overall, this sector alone is responsible for a quarter of the total CO2 equivalent emissions produced by human activity, of which more than half comes from meat production.

The need for cutting and containing emissions is quite evident, as also demonstrated by the European Commission itself by launching the Farm to Fork (F2F) strategy (European Commission 2020), with the goal of turning the eurozone into a model of sustainability also in the food sector. In this respect, this article explores the opportunity of obtaining improvements in the environmental performance of companies by acting on the company's strategy and on the characteristics of the supply chain.

Case study development

The main steps of the *3S Approach* adopted for the development of the case study are the following:

1. Problem setting

- Analysis of top performing players in the yogurt production industry in the Italian market.
- Identification of yogurt supply chain characteristics and constraints for what concerns the Italian market.
- Identification of the supply chain perimeter to be represented in the model.
- Identification of the principal areas of intervention for the enhancing of the sustainability performance.

2. **Data collection:** the model has been created with data coming from different researches and applications (see Annex 1), all considered of high reliability given their academic and/or professional nature. This aspect presents of course a limitation that could be overcome whenever more reliable and precise analysis will be available.

Data used in the model can be clustered as follows:





- Financial data have been collected through the review of several documents and major companies' financial statements.
 - Data about the distribution of customers demand and facilities (suppliers, factories, and warehouses) on the Italian territory have been collected through market research and the analysis of the principal supply chains of dairy industry's major companies.
 - Peculiar data referencing to yogurt production and supply chain have been collected through interviews with industry experts and documents available on the web.
 - Environmental data of each process, such as consumption of resources (water and energy) and harmful emissions (measured in term of equivalent kilos of CO₂) have been gathered through the review of sustainability reports of major producers and with other academic and scientific sources (e.g. emission factors databases property of Intergovernmental Panel on Climate Change of the UN).
3. **Assumptions formulation and validation:** after fetching and preparing the data useful to develop the model, all critical aspects and hypothesis were checked and validated by industry experts. The model was realized with the following assumptions:
- Focus on one finished product: in order to simplify the model and its functioning, it was considered just one type of finished product, a simple white yogurt with no fruit and a standard selling packaging (jar of 125 grams).
 - Perimeter limited to Italian market: the model considered the portion of the supply chain serving uniquely the Italian demand of yogurt. Therefore, no export flows to other countries were considered.
 - Supply chain perimeter: model's boundaries were framed as follows: in the upstream processes, the starting point was set after the first activities of milk processing. Therefore, all the activities of farming and pre-processing were not considered (even because not in direct control of the yogurt producing company). For downstream activities, the end of the supply chain was set at the level of the distribution warehouse of the retailers/GDO players/etc. For this reason, the impacts of product consumption and disposal were not considered.
4. **Model development:** the development of the actual situation and of the possible future scenarios has been carried out with the simulation software AnyLogistix®.
5. **Results analysis and validation:** results were critically discussed and reviewed with industry experts. After some rounds of fine tuning, the model and results were validated by experts and considered of high reliability.



Modelling the actual supply chain

The model of the supply chain has been built based on examples coming from the most important companies of the Italian dairy sector, whose key features have been appropriately resumed and adapted to the simulation context.

A logical picture of the finalized digital twin is reported in figure 1.

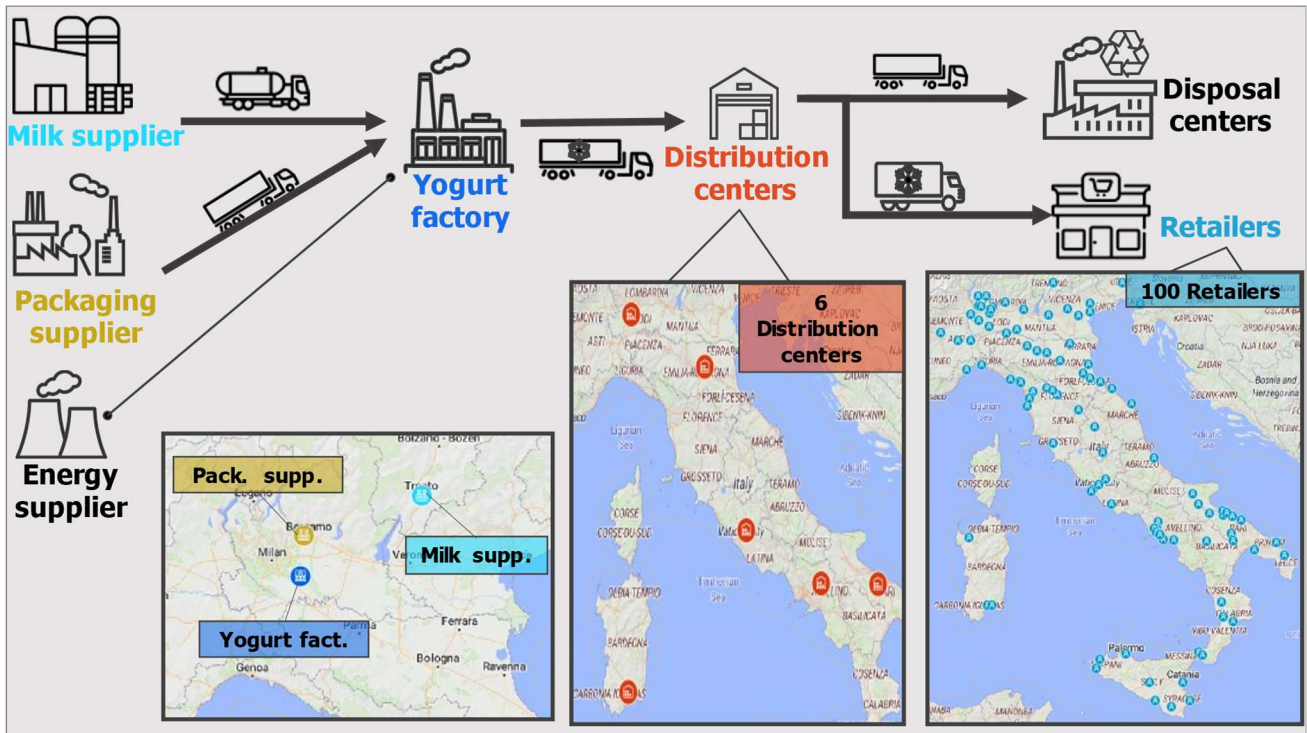


Figure 1: High level scheme of the yogurt supply chain represented in the simulation model.

Entering into details, the major characteristics of the modeled supply chain are the following:

- Upstream in the production chain, two suppliers have been included: one in charge of supplying the raw materials (milk and powdered milk) used to produce yogurt, while the second supplies the material useful for primary packaging (jars), secondary packaging (cartons/trays), and tertiary packaging (plastic film for palletization). Both suppliers are located in North Italy.
- The distribution network considered is made of one-tier (up to the retailers) constituted by six distribution centers which serve the Italian territory. They receive the pallets of finished packaged yogurt coming from the yogurt factory and consequently supply the retailers warehouses on the national territory.
- Downstream, in the supply chain the final customers (retailers) receive pallets with packaged yogurt, ready to be sent to the supermarkets to be sold.
- Finally, disposal centers have been included for managing the finished products that cannot reach the shelves, for several reasons (expiration date, damage during transportation/handling, etc.).

-

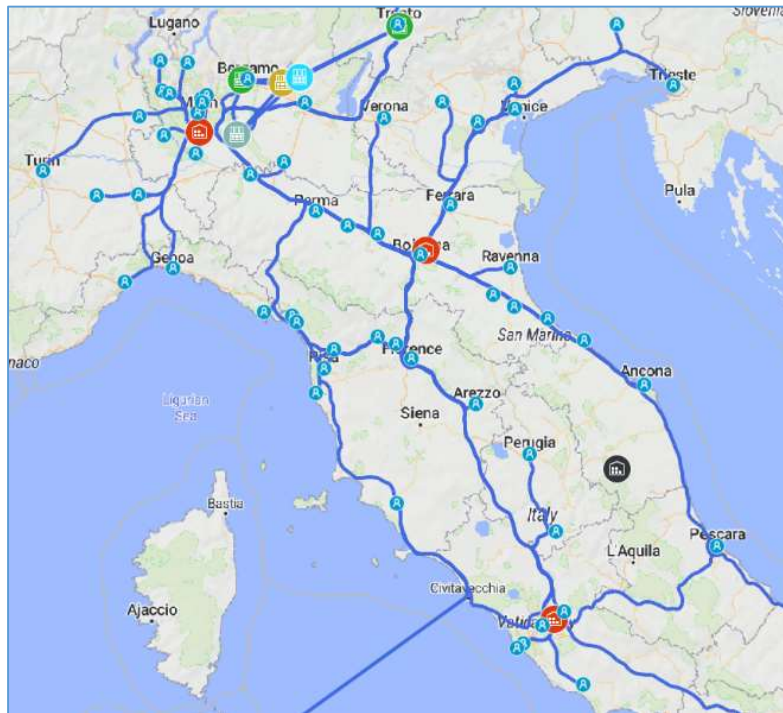


Figure 2: Geographic representation of the supply chain, showing connections between different nodes. Credits AnyLogistix ®.

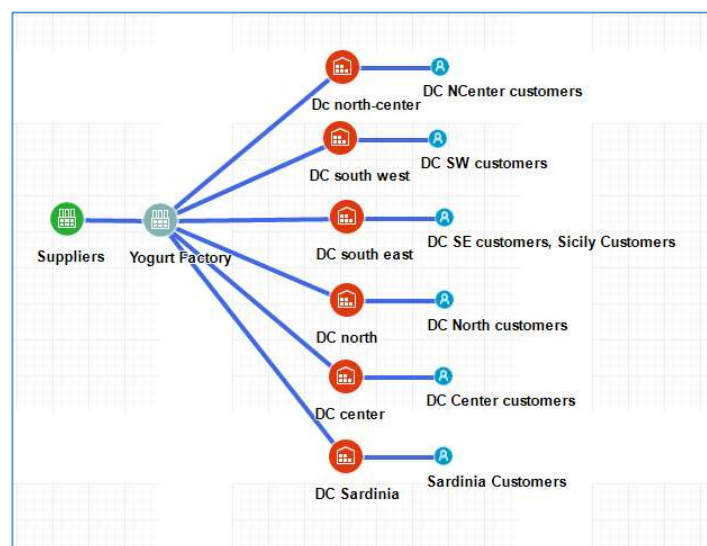


Figure 3: Logical network of the supply chain model. Credits AnyLogistix®.



Identifying the directions of improvement

The simulations carried out on the described supply chain have allowed the possibility to frame its carbon footprint, by dividing it into its fundamental components (figure 4).

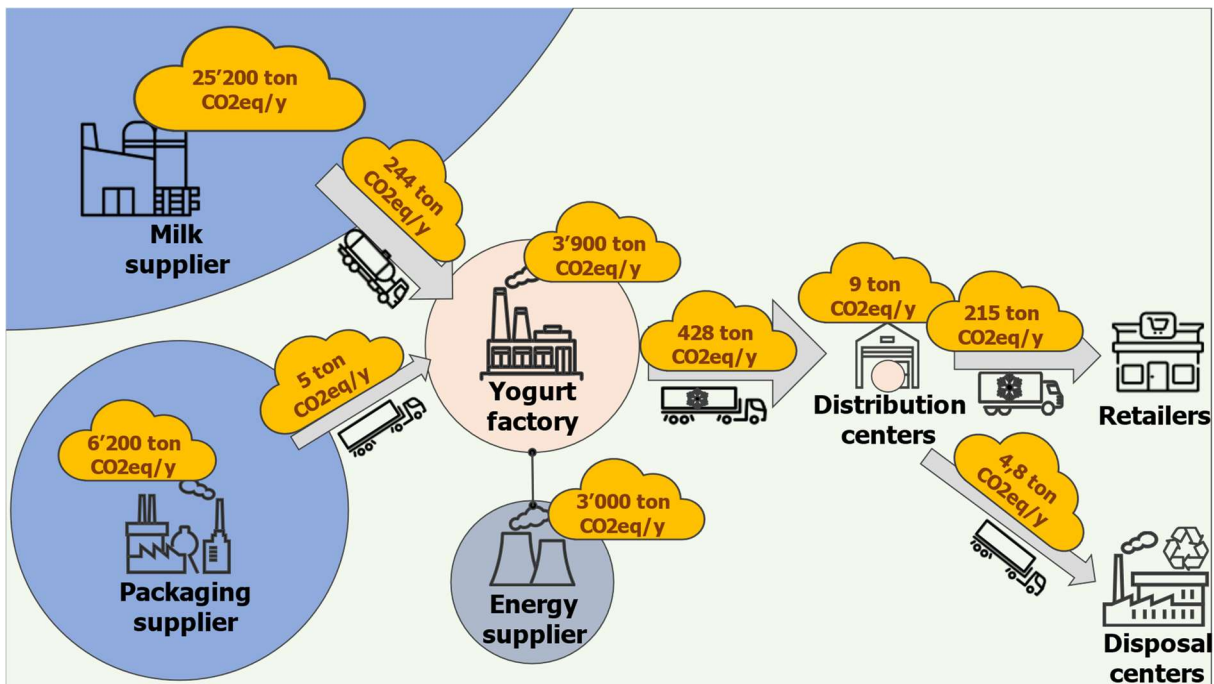


Figure 4: Decomposition of supply chain carbon footprint by facility and transportation path.

The analysis of the results highlighted some aspects which have led to investigate paths for improvement, guiding the company to achieve a more sustainable performance.

The main criticalities of the actual supply chain, for what concerns the environmental performance, were:

- **Generation of waste**, mainly attributable to:
 - physiological scrap production during the yogurt production process, equal to 6,4% of total yogurt production (in weight)
 - expired yogurt across the distribution network, equal to about 12% of total production.
- **Relevant carbon footprint of the supply chain** analyzed, especially concentrated in the upstream activities of the production process (milk processing and packaging manufacturing).
- **Emissions from logistic and distribution** accounts up to 10% of the direct emissions for whom the yogurt producer is accountable (e.g., yogurt production and distribution). This identifies an opportunity for reducing emissions by operating on transportation policies (e.g., enhanced consolidation).



Building the sustainability strategy

Starting from the previous facts about As Is performance and relying on information coming from top industry standards and innovation, several sustainability-oriented strategies have been identified and selected for a following test, via simulation.

The improvement strategies have been selected with the objective of streamlining the supply chain, replacing the materials used for primary and secondary packaging and implementing interventions in distribution that could reduce the generation of waste and the consumption of energy, water and raw materials.

It is not in the aim of this research to investigate all the potential actions that could be taken in respect of supply chain sustainability, since they largely depend on company size, competition market, geography and so on.

For this reason, “quick win” actions such as changing gradually the vehicle fleet in favor of more sustainable truck models were discarded from the choice of the potential actions, depending too much on company characteristics (e.g. large multinationals usually have their own fleet while medium-small companies often rely on vehicles provided by one or more LSPs and hence have no visibility and power on this lever).

At the end of the selection process, **packaging and distribution** network have been targeted as the principal areas of intervention, as confirmed by several sources review. With reference to packaging, the following initiatives have been considered:

- **Primary packaging** change: two alternative scenarios have been hypothesized, involving the replacement of plastic jars with paper or glass jars. After the first simulation runs, the alternative of glass jars was discarded since all traditional and environmental performances were negatively affected.
- **Secondary packaging** change: replacement of the cardboard tray, containing three packages of finished yogurt, with a cardboard box capable to contain six packages of finished yogurt (stacked).
- **Tertiary packaging** change: use of an innovative spray glue for fixing cardboard boxes on the pallets, which completely replaces plastic filming.

Regarding the distribution network, the actions that have been evaluated concerned the **centralization of network stock at factory level** instead of at distribution center level. This activity should have reduced the volumes of expired product (waste) while maintaining a robust service level. In this hypothesis, cost related to storage capacity expansion in the factory were considered as well as cost for closing some of the distribution centers, due to dismantling activities.

For each of the aforementioned strategies relevant data (e.g., emissions from new packaging production) have been investigated and included in the simulation model leading to the creation of 3 scenarios, differing for the span of the improvements of the supply chain considered:



1. **Light impact scenario:** implementation of packaging changes concerning tertiary and secondary packaging (during the first year of simulation) and consequently of primary packaging changes (starting in the second year and becoming effective at the beginning of the third).
2. **Medium impact scenario:** same changes of the previous scenarios with the addition of new distribution policies: customers near the factory will be served directly from it, without relying on local distribution center. Due to geographic dispersion of the customers, only a restricted number resulted to be reachable with this solution (only 7 out of 100 customers were in the service range).
3. **High impact scenario:** same changes concerning packaging (Light scenario) with the addition of the re-design of the supply chain distribution network. In this scenario the total number of distribution centers (DC) is reduced from 6 to 2: two strategic warehouses are kept on the center and southern part of the country, while the factory itself will serve as distribution center for the northern part of the Italian market. More specifically, a preliminary center-of-gravity analysis has been conducted in order to determine the optimal location for establishing the two warehouses: the results show that for the DC in the center of Italy, located near Rome, the location is still optimal, therefore it will be kept and expanded in capacity. While for the DC in the south, it will be moved from its original location (near Benevento, Campania Region) to another place not so far away, in Sorrento. This change in the network entails increased cost for the new facilities or for the capacity increase as well as consistent closure costs for the DCs that were not kept in the new configuration (about 1 mln€ of costs related to the shutdown of the old facilities).

Each scenario was then simulated over a 5-year period, with the aim of deriving the possible impacts on the top line of the P&L and, of course, on the sustainability performance of the company.

A synthesis of the results obtained is reported below:

Performance (cumulated on 5 years)	As Is	Light Scenario			Medium Scenario		Heavy Scenario	
EBITDA	222,3 mln	230,5 mln	+3,7 %		226,1 mln	+1,7 %	233,6 mln	+5,1 %
CO2eq emissions (overall SC)	196.489 ton	184.153 ton	-6,3 %		184.280 ton	-6,2 %	174.116 ton	-11,4 %
Expired finished yogurt	8.450 ton	9.643 ton	+14,1 %		9.694 ton	+14,7 %	5.224 ton	-38,2 %
Plastics used	3.569 ton	717 ton	-79,9 %		714 ton	-80,0 %	692 ton	-80,6 %
Cardboard used	2.083 ton	2.737 ton	+31,4 %		2.745 ton	+31,8 %	2.583 ton	+24,0 %

Table 1: Principal performance indicators of the simulated scenarios.





As it clearly appears, all the three scenarios show a general improvement in major performance such as carbon footprint and profitability.

However, there are also specific areas of performance that show depletions such as for example the volume of expired (wasted) yogurt.

This is a typical decisional situation in which trade-offs between key performance indicators and success factors need to be carefully evaluated by top managers and directors in order to implement actions that are aligned with the company's strategic direction. Under these circumstances, the simulation model is used to provide quantitative estimations to decision makers to support the undertaking of a set of actions which constitute in the end the sustainability strategy for the company's supply chain.

In addition, the selection of the optimal scenario to pursue should be made by considering the company's strategic objectives and resource availability:

- Light impact scenario allows to reach a good reduction in emissions, while strengthening the financial performance, without affecting the structure of the supply chain. The focal point in this situation is the redesign of packaging, which entails a process of negotiation and co-design with the packaging supplier. In a nutshell, this alternative enables a good sustainability performance without affecting the actual organization of the company. On the other hand, this scenario deteriorates the performance of waste generation, which could be a critical success factor for the company's stakeholders.
- Medium impact scenario, on the contrary, seems quite ineffective if compared to the light impact one. Both financial and environmental performance are dominated by the results achieved with the light impact scenario, therefore there is no evident reason for not eliminating this alternative among the three possibilities.
- High impact scenario, finally, seems the most promising one in terms of financial and environmental returns. The enhancements registered in all performance areas, however, require a great change in the organizational and physical structure of the supply chain: for this reason, such choice is not always possible to be implemented in all companies since it requires lot of time, resources (financial and human capital) and commitment.

In order to guide the choice of a scenario, it is possible to additionally condense the results obtained in a **synthetic indicator** capable of describing, in a single picture, how the various scenarios behave. In this situation a ratio between overall emissions and generated value (EBITDA) was chosen, since this indicator is capable of describing the "cost" sustained by the environment (in term of emissions) for each euro of value generated by the company.

The indicator can then be represented as shown in the following graph, depicting a very clear picture of the scenarios analyzed: among the alternatives, the "Heavy impact scenario" is the most performing one, **demonstrating a potential reduction of the emissions on EBITDA indicator of about 16%.**

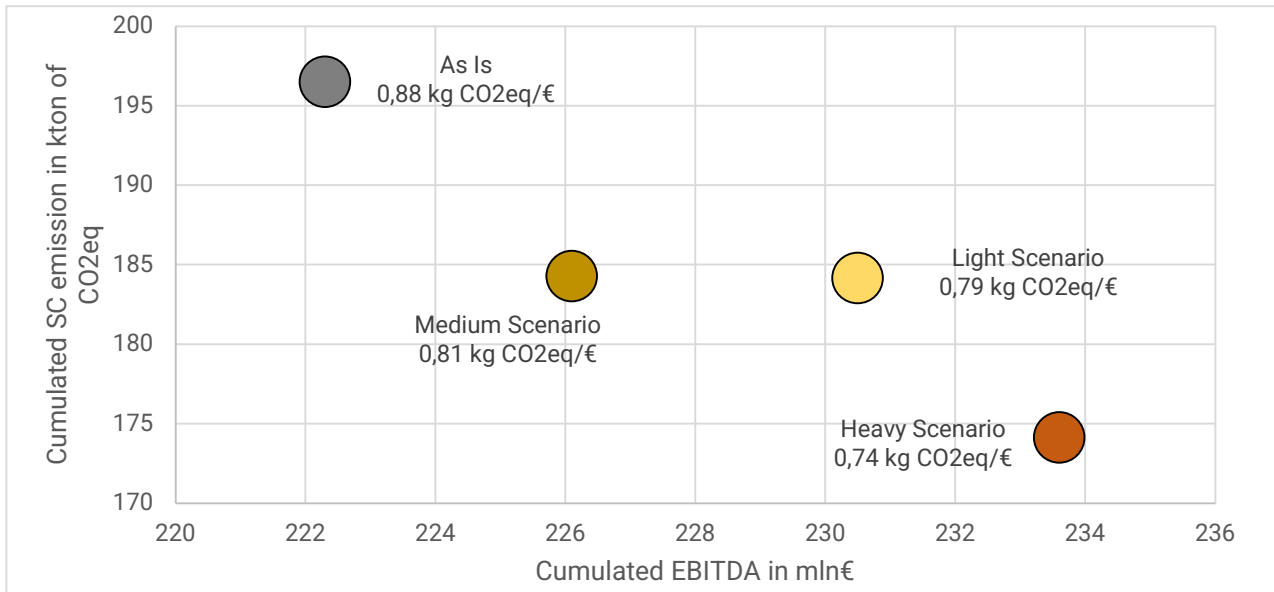


Figure 5: Synthetic performance indicator of the simulated scenarios.

Of course the decision about the strategy to implement (reflected by the three scenarios) cannot be made just on synthetic indicators, but need further investigation on the critical performances.

For this reason, in the hypothesis to select the heavy scenario as a possible direction for improvement, a review of its major performance has been conducted.

First of all, it is necessary to focus on the changes applied to the distribution network, which modifies the logical scheme of the supply chain as follow:

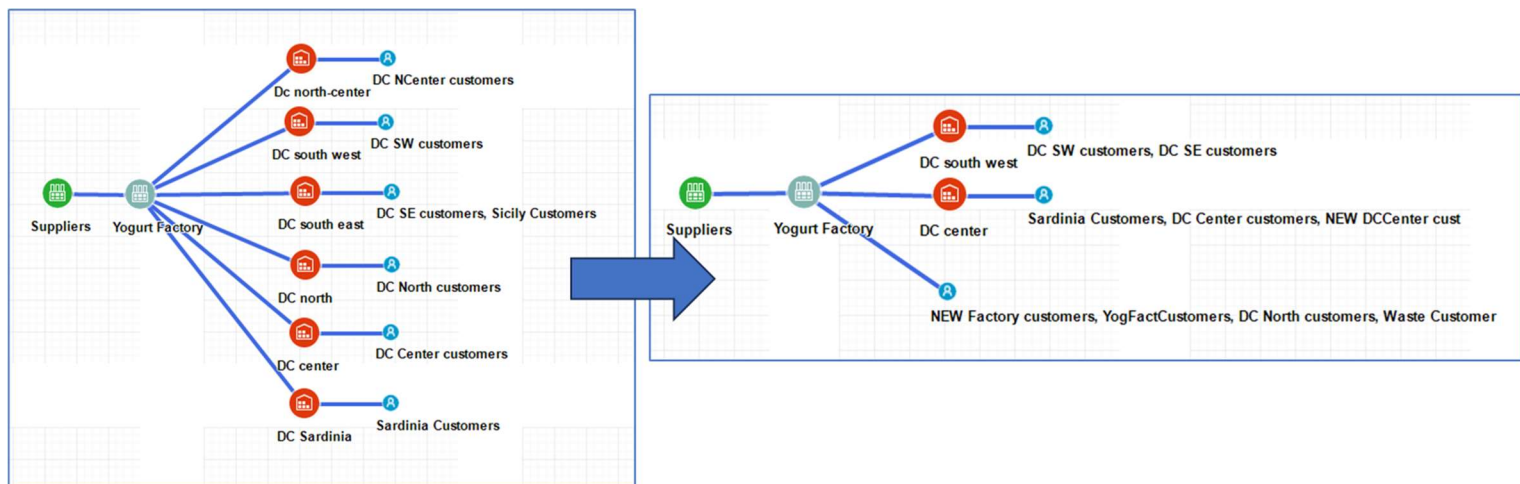


Figure 6: Evolution of the supply chain network from the As Is scenario to the Heavy impact one.
Credits AnyLogistix ®.



For what concerns the financial and operational performance, since revenues are not affected by the changes in the scenario, the major difference registered are related to the logistic and warehousing activities, as it clearly emerges from the table:

Cost item (mln €), cumulated on 5 years	As Is	Heavy Scenario	
Packaging cost	19,2	16,4	-14,9 %
Transportation costs	35,4	30,1	-15 %
Warehousing costs	10,8	15,8	46,7 %
Purchasing costs (raw materials and energy)	54,9	50,4	-8 %
Manufacturing costs	29,4	25,2	-14,5 %
Closing cost		1,0	100 %
Total supply chain cost	149,9	138,9	-7,3 %

Table 2: Detailed benchmark of supply chain costs among the two scenarios.

Table 2 shows how the efficiency derived from the new distribution network is partially counterbalanced by the required increase of stocking capacity in the distribution centers and in the factory. Nevertheless, the **overall reduction of supply chain costs is quite significant**, also considering that this objective is expected to be reached in the medium term.

Shifting the focus to the environmental performance, which constitutes the real objective of the analysis, a detailed examination of each performance factor has been conducted. First of all, the re-design of the distribution network carried a **great improvement on the waste performance**: the reduction of distribution centers implied a consolidation of stock and consequently a reduction in the volumes of expired yogurt. Expired finished products is unfortunately a physiological characteristic of every fresh good industry: since the stock policies are set to maintain a certain service level (often close to 99% in order to deal with large-scale retailers and supermarkets), a surplus in the stocking locations is always present, entailing the possibility for the goods to reach the useful expiration date before reaching the shelves. Under this circumstance, limiting at maximum the volume of expired products is not only beneficial from an economic point of view but also from an environmental one. The proposed heavy scenario helps to **reduce the waste of about 38%**, significantly improving the sustainability performance of the company. This result, already surprising, additionally leads to other benefits, reachable thanks to the systemic nature of the supply chain considered. In fact, the reduction in the downstream level of inventories produces a **waterfall effect which propagates upstream in the supply chain**: fewer expired products mean in fact less production and transportation requirements.





While this effect has a good return in term of financial indicators (as reported in the previous table), it has instead an enormous impact in term of emission reduction, as showed in the following picture.

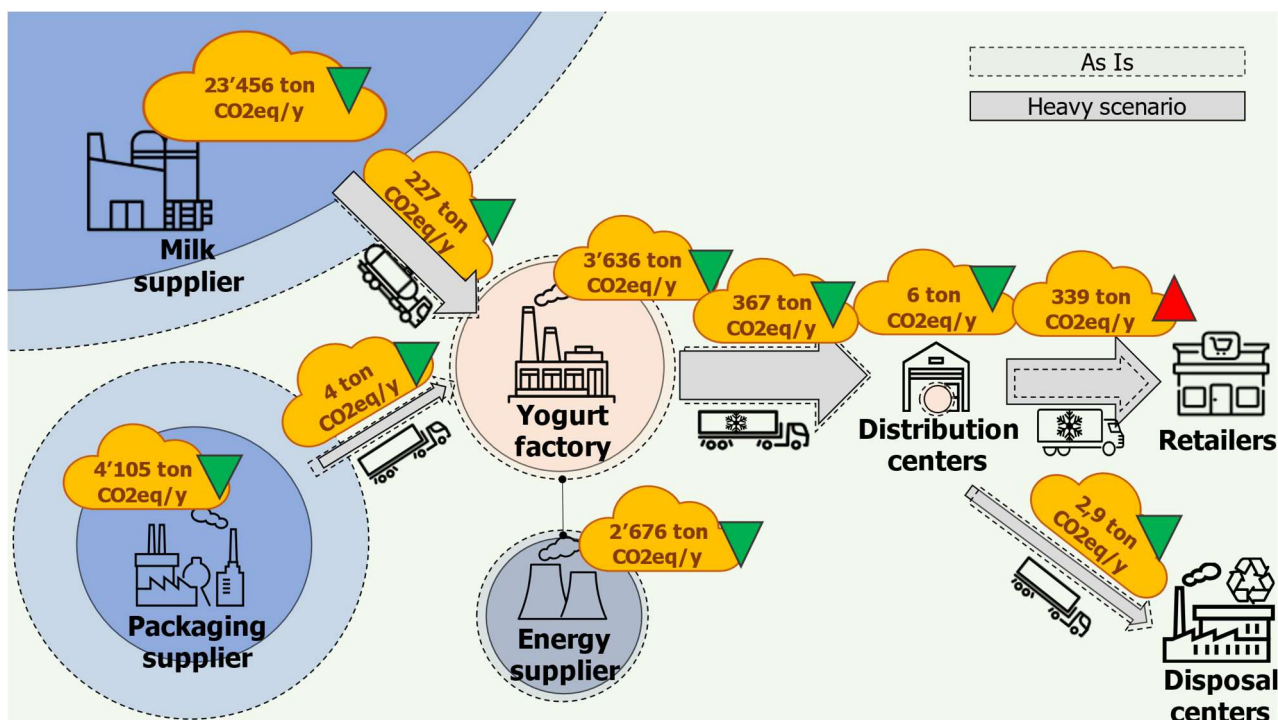


Figure 7: Decomposition of supply chain carbon footprint by facility and transportation path in the Heavy impact scenario.

As it clearly appears, the reduction of final inventories produces a positive effect on emission restraint for every step of the supply chain apart from the final delivery segment (from distribution centers to retailers), which worsen its performance due to the major km travelled caused by the reduction of distribution centers.

All the other stages register a diminution in their emissions, independently from the nature of the activity performed (transportation, production, warehousing). Among them, the most impacting is for sure the production of milk, with **a net saving of around 1.740 ton of CO2 equivalent every year**. This great result demonstrates how a careful investigation and study of the actual supply chain, with the aid of advanced quantitative methodologies such as simulation, could lead to define the direction of improvement which also involves areas and actors not directly under the control of the company (such as the suppliers of milk and energy).

The second area of intervention, concerning primary, secondary and tertiary packaging, has produced other significant results: emissions from packaging production have decreased by more than 30%, meaning a saving of around **2.000 tons of CO2 equivalent every year**. Once again, the effect produced by the change in the materials (cardboard instead of plastic) has been amplified by the general reduction of packaging requirements coming from the resizing of stock levels.





Conclusions

The proposed case study highlights several methodological insights which the author recognizes to be of great importance for every kind of reader, independently of the business area (academic, professional, etc.). The main highlights could be summarized as follows:

1. **Sustainability belongs to the set of company's key performance** and therefore should be treated in a serious and analytical manner. Actions undertaken in the field of sustainability without considering the potential effects and/or criticalities on the supply chain may result in wrong decisions and performance deterioration.
2. **There is not a uniquely recognized approach to deal with sustainability**, nor there are formulas for computing target emissions. This subject should be carefully treated and inserted among top managers agendas and needs to be handled with analytical and systemic thinking in order to be correctly implemented. Under this view, simulation modelling emerges as one of the leading technologies to deal with this kind of projects.
3. **Sustainability is not a linear performance** but depends on the complexity and structure of the company's supply chain. Therefore, small actions implemented at the boundaries of the supply chain may generate great results or worsen the performance instead.
Understanding ex ante the possible implications of modifications in the supply chain is then crucial to act for a positive change of both traditional and environmental performances.
4. **Actions for increasing the sustainable performance should be organized collectively in a long-term strategy**, which may or not constitute a severe investment for the company (as represented in the scenarios proposed) according to its specific objectives. Since most of the time every action has its positive and negative drawbacks, combining actions across a sustainable strategy is a good way for counterbalancing the disadvantages and amplifying the returns.

Entering in the case study, the improvements achievable with the formulation of the scenarios proposed cannot be claimed to be valid globally as none of the proposed strategies is in fact to be considered as the only viable solution. On the contrary, it is possible to demonstrate how decisions taken in the field of sustainability must be often accompanied by a careful analysis of all the others performance areas impacted. In this framework, the role of simulation clearly emerges as the most suitable approach for this type of evaluation.

The purposes of this paper are therefore exploratory, descriptive and explanatory with respect to the sustainability theme and the proposed *3S Approach*.





References

- Al-Emran, Mostafa, e Charla Griffy-Brown. «The role of technology adoption in sustainable development: Overview, opportunities, challenges, and future research agendas.» *Technology in Society* (vol 73), 2023.
- Belvedere, Valeria, e Alberto Grando. *Sustainable Operations and Supply Chain Management*. John Wiley & Sons, 2017.
- Bové, Anne-Titia Bové, e Steven Swartz. <https://www.mckinsey.com/>. 2016.
<https://www.mckinsey.com/~media/McKinsey/Business%20Functions/Sustainability/Our%20Insights/Starting%20at%20the%20source%20sustainability%20in%20the%20supply%20chain/Starting-at-the-source-Sustainability-in-supply-chains.pdf>.
- CBS Statistics Netherlands. s.d. <https://www.cbs.nl/en-gb/news/2022/06/urgenda-reduction-target-for-ghg-emissions-achieved-in-2020/co2-equivalents>.
- du Plessis, M.J., J. van Eeden, e L.L. Goedhals-Gerber. «The Carbon Footprint of Fruit Storage: A Case Study of the Energy and Emission Intensity of Cold Stores.» *MDPI Sustainability*, 2022.
- European Commission. «European Commission website.» 2020. https://food.ec.europa.eu/system/files/2020-05/f2f_action-plan_2020_strategy-info_en.pdf.
- European Environment Agency. June 2023. https://www.eea.europa.eu/data-and-maps/daviz/co2-emission-intensity-13/#tab-chart_4.
- Figge, Frank, e Tobias Hahn. «Is green and profitable sustainable? Assessing the trade-off between economic and environmental aspects.» *International Journal of Production Economics*, 2012.
- Grabolle, Andreas, e Tanja Loitz. *CO2-Zähler*. Pendos, 2007.
- in Naturale*. May 2020. <https://www.innaturale.com/granarolo-presenta-i-nuovi-yogurt-yomo-in-vasetti-di-carta-riciclabili>.
- Ivanon, Dmitry. *Supply chain simulation and optimization with anyLogistix*. Berlin School of Economics and Law, 2018.
- Kumar, Vinod, Zillur Rahman, A.A. Kazmi, and Praveen Goyal. "Evolution of Sustainability as Marketing Strategy: Beginning of New Era." *Procedia - Social and Behavioral Sciences*, volume 37, 2012: 482-489.
- León-Soriano, Raúl, María Jesús Muñoz-Torres, e Ricardo Chalmeta-Rosaleñ. «Methodology for sustainability strategic planning and management.» *Industrial Management & Data Systems*, 2010: 249 - 268.
- Montanino, Andrea, Simona Camerano, Alberto Carriero, Davide Ciferri, e Eleonora Padoan. *La sfida della sostenibilità per la filiera agroalimentare italiana*. CDP Think Thank, 2021.
- Olejnik, Konrad. «Water Consumption in Paper Industry – Reduction Capabilities and the Consequences.» *Security of Industrial Water Supply and Management*, 2011: 113-129.
- Priadi, G., F. Setiyoningrum, e F. Afiati. «The shelf life of yogurt starter and its derivatives based on the microbiological, physical and sensory aspects.» *IOP Conference Series: Earth and Environmental Science* (462), 2020.
- Ries, J. M., E. H. Grosse, e J. Fichtinger. « Environmental impact of warehousing: a scenario analysis for the United States.» *International Journal of Production Research* (55), 2017: 6485-6499.
- Signode. s.d. <https://www.signode.com/en-us/productslist/lock-n-pop/>.
- Singh, Nitish, e Animangsu Ghatak. «Gecko-inspired hierarchically rough surface used as a reprintable paper.» *Chemical Engineering Journal Advances* 12, 2022.
- Sun, Mingxing, Yutao Wang, Lei Shi, e Jiří Jaromír Klemeš. «Uncovering energy use, carbon emissions and environmental burdens of pulp and paper industry: A systematic review and meta-analysis.» *Renewable and Sustainable Energy Reviews*, Volume 92, 2018: 823-833.
- The International EPD System*. October 2021. <https://www.environdec.com/library/epd482>.
- United Carriers APAC Pty Ltd . s.d. <https://unitedcarriers.com/co2-carbon-calculator>.
- Yang, Zhuoqian, James E. Tate, Eleonora Morganti, e Simon P. Shepherd. «Real-world CO2 and NOx emissions from refrigerated vans.» *Science of the Total Environment*, 2021.





Annex 1

Parameter	Value (estimate)	Model application	Source
Carbon emission from primary transportation (WTW computation) – no refrigerated; 30 tons of cargo capacity	0,057222 kg CO ₂ eq/ km / ton of product shipped	Computation of transportation emissions for the flows from packaging supplier to yogurt factory	(United Carriers APAC Pty Ltd s.d.) Computations made on a standard delivery route, comprising both suburban and urban paths. Data about the route considered: • Origin: Milan; Arrival: Rome. Total distance: 480 km • Shipment weight: 30 tonnes • Total CO₂eq emission (WTW): 824 kg
Carbon emission from primary transportation (WTW computation) – refrigerated (0-5 C°); 20 tons of cargo capacity	0,0765625 kg CO ₂ eq/ km / ton of product shipped	Computation of transportation emissions for the flows: 1) from milk processing supplier to yogurt factory 2) from yogurt factory to distribution centers	(United Carriers APAC Pty Ltd s.d.) Computations made on a standard delivery route, comprising both suburban and urban paths. Data about the route considered: • Origin: Milan; Arrival: Rome. Total distance: 480 km • Shipment weight: 20 tonnes • Total CO₂eq emission (WTW): 735 kg
Carbon emission from last mile logistic (WTW computation) – refrigerated (0-5 C°); 4 tons of cargo capacity	0,125 kg CO ₂ /km/ton of product shipped + 0,1675 gr NO _x /km/ton of product shipped Final value: 0,175 kg CO ₂ eq/km/ton of product shipped	Computation of transportation emissions for the flows from distribution centers to retailers warehouses and supermarkets	Refrigerated van emission were derived from the work of (Yang, et al. 2021), by computing an average value among the ones computed according to the different traffic and loading conditions. Conversion factors from NO _x emissions to equivalent CO ₂ were taken from (CBS Statistics Netherlands s.d.).
Carbon emission from warehousing	0,022 kg CO ₂ eq/ ton of stored product/ day	Computation of emissions related to the warehousing of finished yogurt in the distribution centers	(du Plessis, van Eeden e Goedhals-Gerber 2022) (Ries, Grosse e Fichtinger 2017)





Carbon emission and resource consumption from milk production	1,9 kg CO₂eq/ kg yogurt produced 163,5 lt of water / kg of yogurt produced 1549 kWh/ kg of yogurt produced	Computation of emissions and resource consumption for the production of milk (data expressed per kg of yogurt produced)	(Grabolle e Loitz 2007) (The International EPD System 2021)
Carbon emission and resource consumption from yogurt manufacturing	0,3 kg CO₂eq/ kg of yogurt produced 15 lt of water / kg of yogurt produced 0,816 kWh/ kg of yogurt produced	Computation of emissions and resource consumption for the process of transformation of raw materials (water, milk and milk powder) in yogurt	(The International EPD System 2021)
Carbon emission and resource consumption from packaging production – plastic jar	0,3 kg CO₂eq/ kg of yogurt packed (equal to) 0,0375 kg CO₂eq/ jar 35 lt of water / kg of yogurt packed 0,436 kWh/ kg of yogurt packed	Computation of emissions and resource consumption for the process of primary packaging production	(The International EPD System 2021)
Carbon emission and resource consumption from packaging production – cardboard (jar, tray, boxes)	0,196 kg CO₂eq/ kg of yogurt packed 2 lt of water / kg of yogurt packed 0,12 kWh/ kg of yogurt packed	Computation of emissions and resource consumption for the process of primary and secondary packaging production	(Olejnik 2011) (Singh e Ghatak 2022) (Sun, et al. 2018)





Carbon emission and resource consumption from packaging production – plastic filming	0,0063 kg CO ₂ eq/ pallet filmed 4,652 kWh/ pallet filmed (data about water not available)	Computation of emissions and resource consumption for the process of tertiary packaging production	(Signode s.d.)
Carbon emission and resource consumption from packaging production – cohesive spray glue	0,0018 kg CO ₂ eq/ pallet 0,83 kWh/ pallet	Computation of emissions and resource consumption for the process of tertiary packaging production	(Signode s.d.)
Carbon emission from energy production	0,234 kg CO ₂ eq/ kWh	Computation of emissions from electricity production	(European Environment Agency 2023)



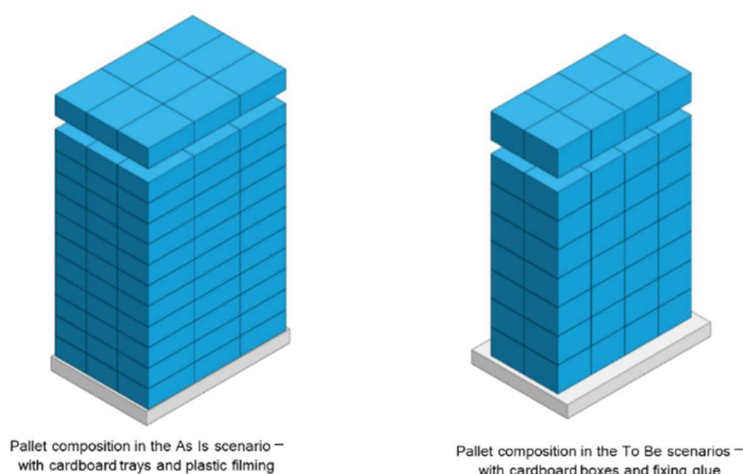


Annex 2

2.1 Volumes and weights

Product	Gross Weight	Dimensions	Finished product content
Plastic jar weight (no filled)	7 gr	Approx. 7*7*7 cm	125 gr
Cardboard jar weight (no filled)	5 gr	Approx. 7*7*7 cm	125 gr
Cardboard tray (no filled)	100 gr	38,5 * 26 * 15 cm	24 jars
Cardboard box (no filled)	180 gr	28 * 28 * 21 cm	48 jars
Pallet EPAL (full), with trays and plastic jar	378 kg	120 * 80 * 180 cm	108 trays on 12 layers, 2.592 jars
Pallet EPAL (full), with trays and cardboard jar	373 kg	120 * 80 * 180 cm	108 trays on 12 layers, 2.592 jars
Pallet EPAL (full), with boxes and plastic jar	442 kg	120 * 80 * 180 cm	64 boxes on 8 layers, 3.072 jars
Pallet EPAL (full), with boxes and cardboard jar	435 kg	120 * 80 * 180 cm	64 boxes on 8 layers, 3.072 jars
Plastic film used for one pallet, with boxes	-	Approx. 18 m2	-
Plastic film used for one pallet, with trays	-	Approx. 17 m2	-
Cohesive glue spray used for one pallet, with boxes	0,05 kg	-	-

Calculations on pallet composition were made through the application <https://onpallet.com/>. The two pallet configurations obtained for the employ of trays or boxes are the following:



2.2 Shelf life

Data	Value	Source
Average supply chain yogurt shelf life – from production up to the positioning in supermarket shelf	10 days	(Priadi, Setiyoningrum e Afiati 2020)

